

THREAT ANALYSIS REPORT

AeroParts Manufacturing Inc.

Manufacturing Sector | Aerospace Components

Date Context: August 2025 Incident

Classification: Confidential

EXECUTIVE SUMMARY

AeroParts Manufacturing Inc. operates in a high-risk cyber environment driven by its role designing and manufacturing aerospace components. As well as its integration of Information Technology (IT) and Operational Technology (OT) systems, also including its reliance on contractors and vendors for production operations. On August 18, 2025, a USB-borne malware incident disrupted Internet of Things (IoT) manufacturing control systems at Plant #2, shutting down two production lines for six hours and causing \$250,000 in delayed production.

This incident exposed weaknesses in endpoint control, identity trust capabilities for contractor access, and network segmentation between IT and OT environments. No confirmed data exfiltration occurred. However, the malware scanned IoT endpoints for vulnerabilities, a behavior consistent with early stage reconnaissance typically conducted by espionage focused threat actors.

According to CrowdStrike's 2025 Global Threat Report, aerospace manufacturers remain prime targets for state-aligned espionage groups, credential abusing cybercriminals, and supply chain intrusion campaigns. These actors increasingly favor stealth, persistence, and credential abuse over noisy, signature heavy attacks. This report evaluates threats and analyzes risk across the seven domains of cybersecurity, maps threat actor Tactics, Techniques, and Procedures (TTPs) to AeroParts specific attack surfaces, and provides prioritized recommendations.

BUSINESS DRIVERS

AeroParts generates revenue through long-term supply contracts with commercial and defense aerospace manufacturers, delivering specialized parts on tight production schedules. The business model depends on three things: uninterrupted production capacity, secure Computer Aided Design/Computer Aided Manufacturing (CAD/CAM) design systems containing proprietary intellectual property, and trusted partnerships with clients, vendors, and contractors.

A disruption to any of these directly impacts revenue. Production downtime delays shipments and triggers Service Level Agreement (SLA) penalties. Intellectual property theft weakens competitive advantage and creates regulatory exposure under the International Traffic in Arms Regulations (ITAR). Losing client trust puts contract renewals and future business at risk.

THREAT LANDSCAPE OVERVIEW

AeroParts faces three primary categories of cyber threat:

- Nation-state actors targeting aerospace intellectual property for strategic and military advantage, with particular interest in CAD designs, material specifications, and defense contract data.
- Criminal groups targeting manufacturers for ransomware and extortion, exploiting the high cost of production downtime and tight delivery schedules that create pressure to pay quickly.
- Supply chain attackers exploit trusted vendor and contractor relationships to bypass perimeter defenses, using compromised partner credentials as an initial access vector into AeroParts systems.

The convergence of IT and OT systems at AeroParts amplifies risk across all three categories. A single compromise on the IT side, such as an infected design workstation, can propagate into manufacturing control systems, turning a contained endpoint incident into a business wide operational disruption. The August 18 incident demonstrated this exact attack path.

SEVEN DOMAINS OF CYBERSECURITY THREAT ANALYSIS

1. Human Domain

Risk Level: Critical

Current Posture: AeroParts employs approximately 600 staff and relies on contract engineers and CAD design contractors with access to sensitive systems. Contractor access is provisioned through sandboxed VLANs with time based accounts.

Gap: The August 18 incident demonstrated that contractor access controls are insufficient. A contract engineer introduced malware via unauthorized USB use on a design workstation. This indicates that physical device policies are either not enforced or not monitored for contractor endpoints. Social engineering and credential abuse remain the most common initial access methods used against aerospace manufacturers.

Threat: Insider threat from both negligent and malicious actors, contractor credential compromise, social engineering campaigns targeting engineering staff with access to proprietary designs.

2. Physical Domain

Risk Level: Moderate

Current Posture: Facilities include two manufacturing plants and one engineering/design center with controlled physical access. The primary data center at HQ uses hardened access controls and monitored server cages.

Gap: Shared workstations at the design center and manufacturing floors provide physical insertion points for removable media. The USB incident originated from a physical device introduced at a design center workstation, indicating that physical security controls do not adequately prevent unauthorized device connections at endpoints.

Threat: Physical introduction of malware via removable media, unauthorized access to industrial control terminals on production floors, tailgating into sensitive areas by contractor personnel.

3. Perimeter Domain

Risk Level: High

Current Posture: Three perimeter firewalls protect HQ and both plants. One Virtual Private Network (VPN) gateway provides role-limited vendor and supplier access with MFA. A vendor portal enables external supplier collaboration.

Gap: While Multi-Factor Authentication (MFA) and firewalls are deployed, credential abuse techniques used by actors like Scattered Spider and Volt Typhoon can bypass these controls through help desk social engineering, MFA fatigue attacks, or session token theft. Prior reconnaissance activity, specifically targeted scans of CAD repositories, indicates that external actors are already profiling AeroParts systems.

Threat: VPN credential compromise enabling unauthorized access, vendor portal exploitation as a pivot point into internal networks, reconnaissance escalation from profiling to active exploitation of identified vulnerabilities.

4. Network Domain

Risk Level: Critical

Current Posture: AeroParts segments networks using Virtual Local Area Networks (VLANs) for OT (IoT), IT (corporate), and guest traffic. Internal VLAN firewalls and Intrusion Detection System/Intrusion Prevention System (IDS/IPS) at the datacenter ingress provide monitoring.

Gap: The August 18 incident proved that segmentation between IT and OT is insufficient. Malware originating on a design center workstation (IT VLAN) was able to reach and flood the IoT control VLAN with malformed packets, shutting down production lines. ERP integration paths between IT and OT create legitimate communication channels that malware can exploit for lateral movement.

Threat: Lateral movement from compromised IT endpoints into OT manufacturing control systems, exploitation of ERP-to-IoT integration pathways, propagation of malware across inadequately segmented VLANs.

5. Endpoint Domain

Risk Level: High

Current Posture: Approximately 600 endpoints including engineering workstations, laptops, and industrial control terminals. A centralized Endpoint Detection and Response (EDR) platform is deployed. USB device monitoring is not yet fully enforced.

Gap: A lack of USB control rollout left the door open for the August 18 incident. Engineering workstations have full CAD system access, making them high value targets. Industrial IoT terminals use weak authentication protocols that were directly exploited during the incident. Behavioral threats and living-off-the-land techniques used by actors like Volt Typhoon reduce the effectiveness of signature-based detection.

Threat: Malware delivery via removable media to engineering endpoints, exploitation of weak authentication on IoT control terminals, use of legitimate tools and credentials to evade endpoint detection.

6. Application Domain

Risk Level: High

Current Posture: Mission-critical applications include CAD/CAM design systems, Data for inventory and scheduling, IoT manufacturing control software, and the vendor collaboration portal.

Gap: Tight integration between CAD and IoT systems means a compromise in one application can cascade into others. The vendor portal creates an externally accessible attack surface. The malware in the August 18 incident specifically targeted vulnerabilities accessible from IoT endpoints, demonstrating that attackers understand and exploit application interdependencies.

Threat: Application-layer attacks targeting ERP integration APIs, vendor portal compromise providing covert access to internal documentation and supply chain data, exploitation of CAD system vulnerabilities for intellectual property theft.

7. Data Domain

Risk Level: Critical (Highest Impact)

Current Posture: AeroParts manages proprietary CAD designs and blueprints, International Traffic in Arms Regulations (ITAR)-controlled defense contract data, and vendor contractual and financial information. Role-Based Access Control (RBAC) with least privilege, MFA for remote access, and a data classification policy restricting cloud storage for design files are in place.

Gap: The quarantining of design systems following the August 18 incident indicates potential exposure of proprietary data, even though no exfiltration was confirmed. Data theft by espionage actors can remain undetected for extended periods. Prior targeted scans of CAD repositories suggest adversary interest in design data specifically. The data classification policy restricts cloud storage, but lateral movement from IT to OT demonstrated that internal access controls have exploitable gaps.

Threat: Exfiltration of proprietary CAD designs and ITAR-controlled data by nation-state actors, long-term undetected data theft causing strategic competitive harm and regulatory penalties, exposure of vendor financial data through portal compromise.

THREAT ACTOR PROFILES RELEVANT TO AEROPARTS

The following profiles map known threat actors to AeroParts-specific assets and attack surfaces.

Actor	Description & TTPs	AeroParts Target	Relevant Domain
Red Apollo (APT10)	Long-term espionage using credential abuse and contractor compromise to steal CAD designs and trade secrets	Contract engineer access paths, sandboxed VLANs, CAD repositories	Human, Data, Network
Volt Typhoon	Targets US critical infrastructure using credential harvesting and living-off-the-land techniques for persistent access without triggering alerts	IT admin credentials, ERP-IoT integration paths, legitimate system tools	Endpoint, Network, Application
AeroBlade	Targets US aerospace organizations with spear phishing and weaponized document payloads for intellectual property (IP) theft	Engineering staff email, CAD file workflows and design documents	Human, Perimeter, Data
TA2541 / Ransomware Groups	Criminal actors using phishing for initial access; ransomware groups (Clap, LockBit, Akira, PLAY) exploit downtime pressure and weak segmentation	ERP systems, production scheduling, IT-OT integration points	Endpoint, Application, Network
Scattered Spider	Social engineering and identity abuse including help desk impersonation to bypass MFA and verification controls	IT help desk, MFA systems, VPN authentication, user credentials	Human, Perimeter
AVIVORE	Targets aerospace supply chains by compromising smaller partners to pivot into larger organizations	Vendor portal, supplier VPN access, third-party contractor accounts	Perimeter, Application, Human

INCIDENT OVERVIEW: AUGUST 18, 2025

Timeline

At 7:50 AM, two production lines at Plant #2 shut down after IoT control terminals lost connectivity. Security alerts identified malformed packets flooding the control VLAN. IT traced the traffic source to a design center workstation infected via unauthorized USB use by a contract engineer.

Attack Chain

- Initial Access: Contract engineer connected an unauthorized USB flash drive to a design center workstation.
- Execution: Malware executed on the workstation and began scanning the local network.
- Lateral Movement: Malware traversed from the IT VLAN to the IoT control VLAN, exploiting weak authentication on IoT devices and insufficient network segmentation.
- Impact: Malformed packets flooded the OT VLAN, causing IoT control terminals to lose connectivity and shutting down two production lines for six hours.

Impact Assessment

- \$250,000 in delayed production costs
- 6-hour production downtime at Plant #2
- Design systems quarantined for forensic analysis
- No confirmed customer data compromise or data exfiltration
- Malware behavior consistent with early-stage reconnaissance by espionage actors

BUSINESS SHUTDOWN SCENARIOS

The following scenarios are ordered by estimated severity of business impact, from highest to lowest.

Scenario 1: Silent CAD Design Exfiltration (Critical)

Likelihood: High | Impact: Critical | Detection Difficulty: Very High

A nation-state actor such as Red Apollo or AeroBlade establishes persistent access through compromised contractor credentials and exfiltrates proprietary CAD designs over weeks or months. The theft goes undetected because no systems are disrupted. AeroParts loses competitive advantage as designs appear in competitor products. ITAR violations trigger regulatory investigation and potential debarment from defense contracts. This is the worst-case scenario because the damage is irreversible and may not be discovered until strategic harm has already occurred.

Scenario 2: Ransomware (Critical)

Likelihood: High | Impact: Critical | Detection Difficulty: Low (post-detonation)

A ransomware group such as LockBit or Akira gains initial access via phishing, moves laterally through inadequate IT-OT segmentation, and encrypts data servers. With the production system down, inventory management, scheduling, and financial operations are paralyzed. Recovery takes days to weeks depending on backup integrity. The tight delivery schedules that drive AeroParts revenue create pressure to pay ransom quickly.

Scenario 3: Contractor Account Compromise (High)

Likelihood: High | Impact: High | Detection Difficulty: High

An actor compromises a contract engineer's credentials through phishing or social engineering. Using legitimate access the attacker escalates privileges or exploits trust to access CAD systems and Production data. Because the access pattern mimics normal contractor behavior, detection depends on behavioral analytics rather than rule based alerts. The August 18 incident demonstrated a version of this scenario with physical rather than credential based access.

Scenario 4: Vendor Portal as Pivot Point (Moderate)

Likelihood: Moderate | Impact: High | Detection Difficulty: Moderate

An attacker compromises a smaller supply chain partner, as AVIVORE has done in other aerospace campaigns, and uses the vendor portal to upload malicious documentation or exploit portal vulnerabilities to pivot into internal AeroParts systems. The vendor portals may receive less security attention than production systems, making it an attractive entry point.

Scenario 5: OT Disruption During Peak Production (Moderate)

Likelihood: Moderate | Impact: Critical (time-dependent) | Detection Difficulty: Low

A targeted attack or malware propagation disrupts IoT manufacturing control systems during a peak production window when SLA deadlines are imminent. The 24-hour maximum tolerable downtime and 8-hour Recovery Time Objective (RTO) for IoT systems mean that even a partial disruption can cascade into missed shipments, SLA breach penalties, and client relationship damage. The August 18 incident was a limited version of this scenario that cost \$250,000 in just six hours.

RECOMMENDATIONS

Recommendations are organized into two tiers: immediate actions addressing vulnerabilities demonstrated by the August 18 incident, and strategic actions addressing broader threat landscape risks.

Tier 1: Immediate (Address Demonstrated Vulnerabilities)

1.1 Enforce Default-Deny USB Policy

Addresses: Human Domain, Physical Domain, Endpoint Domain

Complete the USB control rollout with hardware whitelisting enforced on all endpoints, including contractor workstations. No removable media should be usable without explicit approval. This directly addresses the root cause of the August 18 incident.

1.2 Harden IT-OT Network Segmentation

Addresses: Network Domain

Deploy monitored gateways between IT and OT VLANs with explicit allow-listing for required communication paths. This directly addresses the lateral movement that enabled the August 18 production shutdown.

1.3 Strengthen IoT Device Authentication

Addresses: Endpoint Domain, Application Domain

Replace weak authentication protocols on IoT control terminals with certificate based or hardware token authentication. IoT devices should not accept connections from unauthenticated sources. This addresses the vulnerability exploited during lateral movement in the August 18 incident.

Tier 2: Strategic (Address Broader Threat Landscape)

2.1 Implement Phishing-Resistant MFA and Zero Trust for Contractors

Addresses: Human Domain, Perimeter Domain

Replace standard MFA with phishing resistant methods for all remote access, prioritizing contractor and vendor accounts. Implement continuous verification rather than single-point authentication for contractor sessions. This mitigates credential abuse techniques used by Red Apollo, Scattered Spider, and Volt Typhoon.

2.2 Deploy Behavioral and ATT&CK-Mapped Threat Hunting

Addresses: Endpoint Domain, Network Domain, Data Domain

Enhance the existing platform with behavioral detection capabilities to identify living-off-the-land techniques and credential abuse patterns that evade signature-based detection. Implement proactive threat hunting mapped to MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) techniques used by the profiled threat actors, specifically T1078 (Valid Accounts), T1091 (Replication Through Removable Media), and T1021 (Remote Services).

2.3 Continuous Vendor Access Monitoring

Addresses: Application Domain, Perimeter Domain

Implement real-time monitoring and anomaly detection on vendor portal activity and VPN sessions. Enforce strict time-bound credentials with automatic expiration and re-verification. Audit vendor access logs against expected behavior baselines.

2.4 Regular IT/OT Tabletop Exercises

Addresses: All Domains

Conduct quarterly tabletop exercises simulating the business shutdown scenarios described in this report. Include IT, OT, manufacturing operations, and executive leadership. Test incident response procedures for both IT and IoT systems separately, as response requirements differ significantly.

CONCLUSION

AeroParts Manufacturing Inc. occupies a threat landscape where nation-state espionage, financially motivated ransomware, and supply chain intrusions converge on a single target. The August 18 incident was not a worst-case scenario. It was a warning that demonstrated a viable attack path from contractor endpoint to production floor through gaps in USB enforcement, network segmentation, and IoT authentication.

The most dangerous outcome for AeroParts is not a visible disruption but a silent one: prolonged, undetected exfiltration of proprietary CAD designs and ITAR-controlled data that erodes competitive position and triggers regulatory consequences long after the initial compromise. The profiled threat actors have demonstrated both the capability and intent to execute this type of campaign against aerospace manufacturers.

The recommended actions are designed to close the gaps demonstrated by the incident (Tier 1) while building resilience against the broader threat landscape (Tier 2). Tier 1 recommendations should be completed within 90 days. Tier 2 recommendations should be initiated within 90 days and fully implemented within 12 months. Progress should be tracked through the existing ISO 27001 audit cycle and reported to executive leadership quarterly.

APPENDIX

1. [Executive Summary](#)
2. [Business Drivers](#)
3. [Threat Landscape Overview](#)
4. [Seven Domains of Cybersecurity Threat Analysis](#)
 - 4.1. *Human Domain*
 - 4.2. *Physical Domain*
 - 4.3. *Perimeter Domain*
 - 4.4. *Network Domain*
 - 4.5. *Endpoint Domain*
 - 4.6. *Application Domain*
 - 4.7. *Data Domain*
5. [Threat Actor Profiles](#)
6. [Incident Overview: August 18, 2025](#)
7. [Business Shutdown Scenarios](#)
 - 7.1. *Silent CAD Design Exfiltration (Critical)*
 - 7.2. *Ransomware (Critical)*
 - 7.3. *Contractor Account Compromise (High)*
 - 7.4. *Vendor Portal as Pivot Point (Moderate)*
 - 7.5. *OT Disruption During Peak Production (Moderate)*
8. [Recommendations](#)
9. [Conclusion](#)

